Fatine Kassabi

Lille, France • fatinekassabi22@gmail.com • +33 7 74 21 63 40 • LinkedIn • github.com/FatineKa

Profile

Final-year Master's student in Data Science, after a first Master's in mathematical modeling, with a strong background in probability, Bayesian statistics and machine learning, currently doing thesis research on LLM-based inference. Seeking a PhD in NLP and machine learning, with a particular interest in large language models and the statistical analysis of their behavior.

Education

MSc in Applied Mathematics and Statistics
Track: Statistical and Numerical Engineering (ISN) — Data Science 2025 – 2026
Université de Lille — Faculty of Science and Technology *Villeneuve-d'Ascq, France*

Coursework: statistical learning theory, machine learning, time series, spatial statistics

MSc in Mathematical Modeling and Data Science 2023 – 2025
Abdelmalek Essaâdi University — Faculty of Sciences and Techniques (FST) *Tangier, Morocco*

Coursework: stochastic differential equations, extreme value statistics, functional analysis, numerical analysis of PDEs, mixed models

BSc in Statistical Engineering 2020 – 2023
Abdelmalek Essaâdi University — Faculty of Sciences and Techniques (FST) *Tangier, Morocco*

Research Experience

Master's thesis: LLM-based Chinese idiom inference April 2026 – present
Université de Lille *Villeneuve-d'Ascq, France*

- Developed a Bayesian inference framework using Qwen2.5 to rank approximately 31,000 Chinese idioms from text, combining language-model likelihoods with a frequency-based prior.
- Implemented and validated Metropolis–Hastings MCMC, including informed proposals based on embeddings and edit-distance neighborhoods.
- Studied selection of the number of mixture components through statistical learning theory.

Bayesian inference | Inverse problems | LLM | Learning theory | Model selection | PyTorch

Research internship: spatio-temporal predictive modeling March 2025 – Sept. 2025
International Water Research Institute (IWRI), UM6P *Benguerir, Morocco*

- Built a high-resolution spatio-temporal prediction pipeline (statistical downscaling of precipitation), integrating 27 years of multivariate climate data, trained on GPU with PyTorch.
- Designed and compared several model families (SVM, Random Forest, Vision Transformer, autoencoder) under a rigorous evaluation protocol, assessing predictive performance and generalization.
- Automated data preparation, training and evaluation to guarantee reproducible experiments on large data volumes.

Spatio-temporal data | Deep learning | Vision Transformer | Autoencoder | Time series | GPU

Data Science Projects

Recommendation and decision-support system (RAG architecture) Jan. 2026 – March 2026
Data Science project: university guidance

- Built an end-to-end RAG pipeline: semantic search combining embeddings, a vector database (ChromaDB) and context-augmented generation, with an interactive application (MySQL, Streamlit).

RAG | Embeddings | Vector search | LangChain | ChromaDB | Streamlit

Skills

Languages	French (fluent), English (fluent), Arabic (native)
Mathematics	Probability, Bayesian statistics, stochastic processes, optimization, statistical learning theory, mathematical modeling
Machine & Deep Learning	PyTorch, TensorFlow, Scikit-learn, Transformers, autoencoders, NLP, time series and predictive models
Generative AI	LLMs, embeddings, RAG architectures, vector search, Hugging Face
Programming & tools	Python (NumPy, Pandas, SciPy), R, SQL, LaTeX; Git, reproducible experiments